

# HENOK WEHIBE

London, United Kingdom

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## WORK EXPERIENCE

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MicrometreUK

*Lead/Full Stack Developer*

*2022 - Present*

Architected and built MicroANPR.com, an end-to-end cloud-native and edge Automatic Number Plate Recognition (ANPR) ecosystem. Designed an API-first multi-tenant architecture supporting parking operators, gated communities, fleet management, and retail car parks. Achieved 97%+ accuracy on UK license plates with ISO/IEC 27001, 27701, and 25010 compliant deployments spanning bare-metal servers, desktop operator stations, and mobile edge AI devices.

- **Core API Engine & Asynchronous Backend (FastAPI / PostgreSQL):** Engineered asynchronous Python/FastAPI backend with SQLAlchemy and PostgreSQL for real-time video job processing, resident/visitor permit tracking, structured JSON security audit logging (`SecurityEventLogger`), and multi-format automated reporting (JSON, CSV, PDF, ZIP archives).
- **Computer Vision & Deep Learning Pipeline:** Built real-time video stream recognition engine using OpenCV and TensorFlow Hub deep learning models (Faster R-CNN, OCR) supporting all UK plate formats with character bounding-box extraction, confidence scoring, and automated permit verification.
- **Edge AI Mobile Camera Application (Flutter / TFLite):** Developed Flutter mobile app for real-time on-device vehicle localization and classification directly on raw camera frame buffers using Google ML Kit Vision and custom quantized EfficientNetB0 TFLite models. Integrated Android Scoped Storage (`MediaStore` API) for zero-permission media exports.
- **Cross-Platform Operator Client (Tauri 2 / Rust):** Built cross-platform desktop and mobile operator client using Tauri v2 (Rust + HTML5/Tailwind) featuring strict Rust capability sandboxing (`capabilities/default.json`) and automated cross-compilation for Linux (`.deb`, AppImage), Windows (`.exe` via MinGW/NSIS), and Android (`.apk`).
- **Infrastructure-as-Code & DevSecOps (Ansible / Docker):** Created Ansible playbooks for automated bare-metal host provisioning, SSH hardening, Fail2ban intrusion defense monitoring Docker Nginx logs, non-root container isolation (`APP_UID/APP_GID`), systemd service auto-recovery daemons, and automated disaster recovery (`backup-restore.sh`).
- **Real-Time Notifications & High-Performance Stream Ingestion:** Implemented Go/Gin microservices and Server-Sent Events (SSE) for zero-latency push notifications to client dashboards, utilizing Redis Pub/Sub for broadcasting parking activity updates across distributed instances.

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## Projects

*2020 - Present*

1. **Browser-Based AI Chat (WebLLM)** - Built a 100% client-side AI chat application using WebLLM and WebGPU with zero server dependencies.
  - Implemented WebGPU for hardware-accelerated AI inference directly in the browser.
  - Demonstrated modern JavaScript and AI/ML integration techniques.
2. **Daily Cashier** - Privacy-first sales management system using WebAssembly and SQLite.
  - Built offline-first application with OPFS (Origin Private File System) for persistent client-side storage.
  - Developed with Next.js and SQLite WASM, requiring zero server dependencies.
3. **RAG Quiz System** - Retrieval-Augmented Generation implementation using Ollama.

- Integrated embeddings and language models for intelligent question-answering.
- Built with Python for backend processing and vector storage.

#### 4. DevOps & Infrastructure Projects

- Developed Ansible roles for automated WireGuard VPN deployment and event-driven infrastructure.
- Configured Kubernetes clusters (kubeadm, Microk8s) with Calico/Flannel networking and Nginx Ingress.
- Built custom Debian/Linux distributions using Packer for consistent development environments.
- Set up private Docker Registry and Gitea instances on AWS, Azure, and GCP.
- Implemented Nginx/HAProxy reverse proxy with SSL termination and Fail2ban security.
- Created monitoring systems with Monit, Crontab, and Git for configuration tracking.

Freelance/Self Employed

Full Stack Web Developer

2020 - 2022

- Developed e-commerce websites, landing pages, and custom web applications for clients using React, Next.js, and WordPress.
- Delivered end-to-end solutions including domain registration, hosting configuration, SSL termination, and ongoing maintenance.
- Conducted comprehensive performance, accessibility, and SEO audits, leveraging Cloudflare and Vercel edge networks.

#### TECHNICAL SKILLS

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|---------------------------|-----------------------------------------------------------------------------|
| <b>Languages</b>          | Python, Go, Rust, Dart, JavaScript/TypeScript, SQL, Bash                    |
| <b>Frontend</b>           | React, Next.js, Flutter, HTML5/CSS3, Tailwind CSS, Astro                    |
| <b>Backend</b>            | FastAPI, Gin (Go), Flask, Node.js, REST APIs, WebSockets, SSE               |
| <b>AI/ML &amp; Vision</b> | TensorFlow (Faster R-CNN, TFLite), Google ML Kit, OpenCV, PaddleOCR         |
| <b>Databases</b>          | PostgreSQL, MySQL, Redis, MongoDB, SQLite (WASM)                            |
| <b>DevOps &amp; Cloud</b> | Ansible, Docker, Kubernetes, Nginx, Fail2ban, Systemd, AWS, GCP, Cloudflare |
| <b>Tools &amp; Build</b>  | Tauri, Git, Packer, Vagrant, FFmpeg, Make, MinGW, NSIS                      |

#### CERTIFICATIONS

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| <b>FreeCodeCamp</b> | Responsive Web Design                     |
| <b>FreeCodeCamp</b> | JavaScript Algorithms and Data Structures |

#### EDUCATION

South Thames College

London, UK

**Diploma** Information Technology

**GCSE** Triple Science (Biology, Chemistry, Physics), English, Maths